

RESEARCH EDITION · ALPHA 0.5 · 1 NOVEMBER 2026

xGEO
CORE

The simulation engine under xGEO: its physical models, the sources they rest on, how each is checked, and how to use it as an open-source tool

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BUILD	TESTS	COMPARISONS	ENGINE
Alpha 0.5	64	680 / 681 pass	Godot 4.7

680

of 681 comparisons within tolerance

64

automated tests against independent references

14

physical domains, from orbits to infrared

4

known simplifications, reported, not hidden

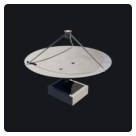
ABSTRACT

xGEO simulates a gray-zone confrontation in low Earth orbit. Its scenario, CLOSE PASS, is built on a real event: in October 2026 the Russian satellite Cosmos 2612 shadowed ICEYE-X36, a commercial radar-imaging satellite, at a few kilometres. The player commands WATCHMAN-1, a notional US inspector, and has to make Cosmos 2612 break off without contact, debris or open conflict, while keeping track of it with real ground radars. The craft start from public SGP4 element sets and fly under the Earth's gravity with its J2 flattening, the Sun and the Moon. This edition sets out every physical model the engine uses, with its equation, its source, how sure we can be of it, and which numbers are *design values* chosen for play. The engine's code is checked against JPL Horizons, hapsira, scipy, astropy, filterpy, ITU-R recommendations and published worked examples: 681 comparisons across 64 tests, 680 within tolerance, and 4 further rows that report known simplifications. Section 11 lists what could not be verified.

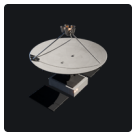
FIG. 1 WHAT THE GAME MODELS, AND WHERE TO READ ABOUT IT



ORBITS & EPHEMERIDES
§2–3 · SGP4, J2, Sun and Moon, Lambert



RADIO & DETECTION
§4 · ITU patterns, Friis, DF



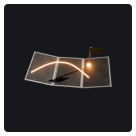
GROUND RADAR
§4.5 · five real sites



HEAT & SIGNATURES
§6 · energy balance, IR, shadow



TRACKING
§7 · unscented Kalman filter, custody



DOCTRINE & SCENARIO
§8–9 · the October 2026 incident

Every model is tested against an independent reference (§10). Click a box to jump to its section.

1 METHOD AND LABELS

The rule is that every number is either sourced or labelled. Each source claim carries one of four labels:

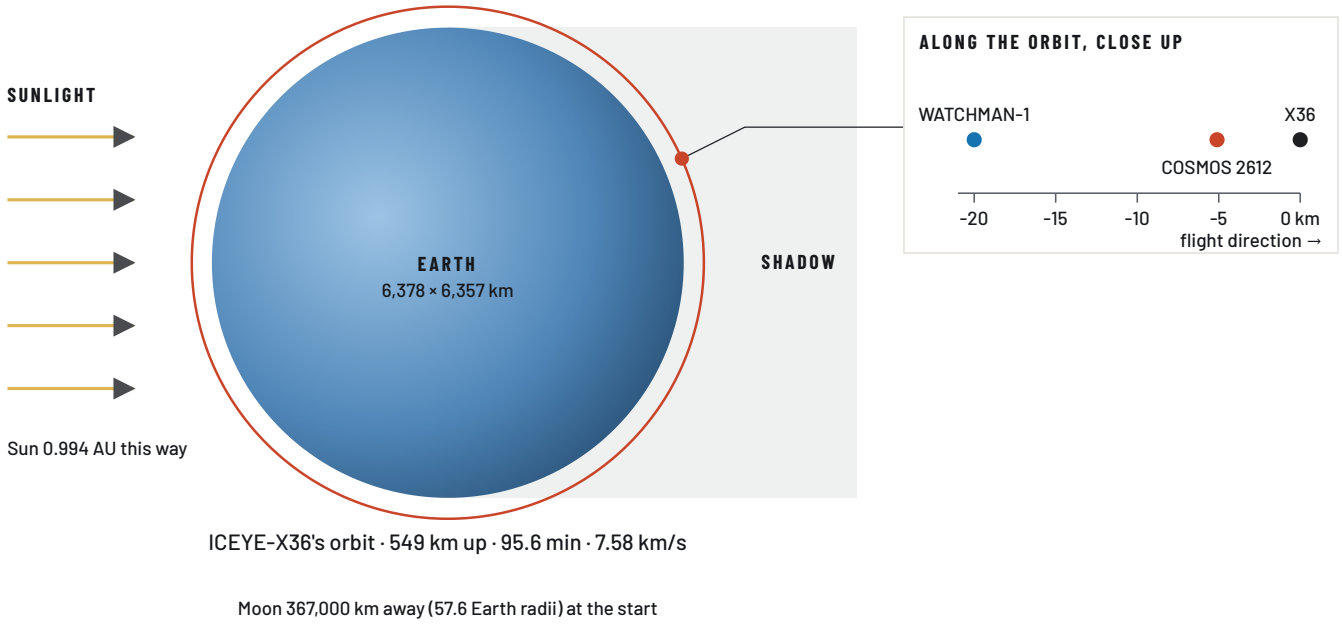
- V-TEXT The document was fetched and the passage read; the section, table or page is given.
- V-RECORD The work exists (publisher, DOI or official listing) but the passage was not read, because of a paywall or because it is a book. The equation used is the standard textbook form.
- UNVERIFIED Not confirmed. Treat it as unsourced.
- DERIVED Computed from verified inputs, with the method stated so it can be rerun.

Values chosen for play carry DESIGN VALUE. A design value is never presented as measured. Where possible, it is chosen inside a range that a source does support. The most important design values in CLOSE PASS are the inspector's plan (where it parks and for how long), the distances at which it gives way or leaves, the three hours of lost contact after which it withdraws (only with a Blue craft within 25 km of it), the tension costs, and the scenario's two days, against the weeks the real event took.

The game runs in Godot 4.7 (GDScript, with 64-bit arithmetic wherever precision matters). Its physics is tested by an external suite (§10), which reads the engine's own outputs. Python never reimplements the game in order to grade it.

2 FRAMES, TIME AND STARTING STATES

FIG. 2 THE SCENARIO, TO SCALE



The Earth and ICEYE-X36's orbit drawn to the same scale, with sunlight from the left and the shadow behind. At this scale the three craft sit on one point; the inset spreads them out along the orbit. Start positions from §2.2.

2.1 Reference frame and time

CLOSE PASS is centred on the Earth. Its axes are those of GCRF, the Earth-centred frame aligned with the ICRF: the Earth's mean equator and equinox of J2000 to within the frame bias of a few milliarcseconds. The Earth is an oblate body, 6,378.137 × 6,356.752 km (WGS 84), with GM = 398,600.4418 $km^3\ s^{-2}$ and J2 = $1.08262668 \times 10^{-3}$ (EGM2008, as tabulated by Vallado[18]). The 3-D view uses a floating origin at the camera's focus, so positions keep their precision at any distance. The clock is UTC, starting at 21:00 UTC on 26 October 2026 and running two days. Leap seconds, TT − UTC and UT1 − UTC are ignored; §2.4 gives what that costs.

2.2 Starting states from public element sets

ICEYE-X36 (NORAD 59103, 2024-043F) starts on its real orbit. Its CelesTrak GP element set (epoch 26 October 2026, 21:06 UTC)[166] is propagated by SGP4 (Vallado et al. 2006[167]), whose output is in the TEME frame, and then rotated into GCRF. The result is a near-circular orbit with semi-major axis 6,924.4 km: 549 km up, a

period of 95.57 minutes and a speed of 7.584 km/s. SGP4 is used only for this starting state. From then on every craft is integrated under the force model of §3, which is how a real analyst would turn a catalogue orbit into a precise one.

Cosmos 2612 (NORAD 68762, 2026-083H) starts 5.1 km behind ICEYE-X36 and 140 m above it, where public element sets put it at 07:38 UTC on 18 October 2026 (KeepTrack's account of the rendezvous[168]). It is placed on the path that does not drift away along-track, which is how an inspector holds station. Blue starts with Cosmos 2612's catalogued orbit, about 200 m off its true one (a design value). WATCHMAN-1, the notional US inspector, starts 20 km behind ICEYE-X36.

CRAFT	ROLE	MASS	ΔV	SENSORS AND RADIO	SOURCE
ICEYE-X36	Target: commercial X-band SAR, operated by ICEYE US	90 kg	none modelled	3.25 m SAR antenna at 9.65 GHz	ESA Earth Online[160]; Gunter's Space Page[169]; GCAT[170]
Cosmos 2612	Red inspector, Russian Aerospace Forces	500 kg	≈ 90 m/s	0.1 m infrared camera; S-band command radio, 5 W into a 0.3 m dish	Mass, engine and radio notional; its observed moves, about 67 m/s from 30 Sep to 18 Oct, from public element sets[168]
WATCHMAN-1	Blue inspector (notional, after GSSAP)	675 kg	259 m/s	0.3 m telescope; 0.5 m receiving dish with a 2 m interferometer baseline	Mass within the 650–700 kg reported for GSSAP[150]; 4 × 22 N hydrazine at 2.2 km/s exhaust speed, 75 kg of propellant: design values

Table 1. The three craft. Values marked notional are DESIGN VALUES; GSSAP's sizes were never published.

2.3 The Sun and the Moon

The Sun's position comes from JPL's approximate Keplerian elements for the Earth–Moon barycentre (Standish & Williams[2], Table 1, valid 1800–2050), reversed to give the Sun as seen from the Earth: 0.9938 AU at the start. The barycentre lies about 4,700 km from the Earth's centre, which turns the Sun's direction by under 7 arcseconds. The Moon comes from the low-precision lunar series of Montenbruck & Gill[20] (§3.3.2, eqs. 3.47–3.48), good to a few arcminutes, about 500 km at the Moon's distance. It is 367,065 km away at the start. GM values come from DE440[3] (the Moon's 4,902.800118 $km^3\ s^{-2}$). Planets are added only beyond the Earth's sphere of influence, where their pull starts to matter.

2.4 The turning Earth and its ground sites

The ground radars of §4.5 and Cosmos 2612's handler near Moscow sit on the turning Earth. A site's position is rotated from the Earth-fixed frame into GCRF by Greenwich mean sidereal time (GMST 1982, Vallado[18] Alg. 15) and by IAU 1976 precession (Alg. 23), at the Earth's rate of 7.292115×10^{-5} rad/s. Nutation (under 20″), polar motion (under 0.5″) and UT1 − UTC (under 0.9 s) are omitted, so a site sits within about 1 km of where a full IERS reduction would put it. A radar measures what it sees from that point; a 1 km site error is about a thousandth of a typical range, and it is shared by every look from that site.

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3 GRAVITY AND TRAJECTORY DESIGN

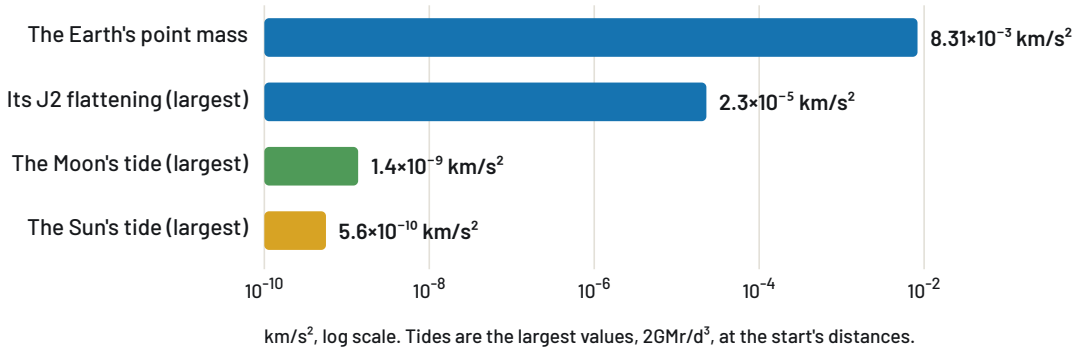
3.1 Force model and integration

The acceleration on a craft is the Earth's point mass, its J2 zonal term, and the Sun and the Moon as third bodies (direct minus indirect term; Montenbruck & Gill[20] §3.3, Cowell's method):

$$\mathbf{a} = -\mu\,\mathbf{r}/|\mathbf{r}|^3 + \mathbf{a}_{J2}(\mathbf{r}) + \Sigma_k\,\mu_k\left[\,(\mathbf{r}_k-\mathbf{r})/|\mathbf{r}_k-\mathbf{r}|^3 - \mathbf{r}_k/|\mathbf{r}_k|^3\,\right]$$

Integration is DOP853, the Dormand–Prince 8(5,3) pair with error control (Hairer, Nørsett & Wanner 1993), every step held to a relative 10^{-12}, so steps lengthen on quiet arcs and shorten near a close pass, with no step rule to tune. Against scipy's DOP853 at 10^{-13} on the same forces, the engine stays within 2.8 mm over a day, and within 0.13 m over a 20-day J2 arc. The classical fourth-order Runge–Kutta step stays in the code for tests. There is no atmospheric drag (§11). The same engine, run on its own, centres the world on Jupiter with the four Galilean moons as its third bodies; the external suite of §10 checks that world against JPL Horizons.

FIG. 3 WHAT PULLS ON A SATELLITE AT 549 KM



The force model of §3.1 at ICEYE-X36's radius. The Sun and the Moon are a millionth of J2, but over two days they move a craft by tens of metres, which is the scale of a close approach.

For the Earth, two checks run in Godot on every change. ICEYE-X36's orbital node turns 1.0395° per day, against 1.0400° from J2's first-order secular rate (Vallado eqs. 9-41/42). And a craft placed on the no-drift path beside it stays within 779 m of it over a whole orbit, which shows that the starting states and the forces agree.

3.2 Two-body tools

Kepler propagation uses universal variables (Curtis[17] Alg. 3.3–3.4), with Danby's starter near e = 1 and Vallado's starter for long hyperbolic coasts (Vallado[18] Alg. 8). Lambert's problem is solved by Izzo's method (Izzo 2015[23]), including multi-revolution transfers. Across 111 test transfers it agrees with hapsira's Izzo solver to 7.3e-15, and it reproduces Curtis Example 5.2. An earlier build used Curtis Alg. 5.2; a loss of precision in its Stumpff function as $z \to 4\pi^2$ made every transfer over 180° fail, which was one reason for the change.

3.3 Refining under full gravity

Over a 3-hour coast in low orbit, J2 alone moves a craft by kilometres, so a two-body Lambert solution misses a target only a few kilometres away. Every planned burn is therefore refined by differential-correction shooting through the full-gravity coast (Vallado[18]; single shooting as set out in NASA/TP-20220014814 §5.2.1). Newton's method starts from a two-body Jacobian and updates it with Broyden's method[24]. A 3-hour move by WATCHMAN-1 from 20 km behind ICEYE-X36 to beside it is planned at 0.74 m/s and flies at 0.75 m/s.

3.4 The planner

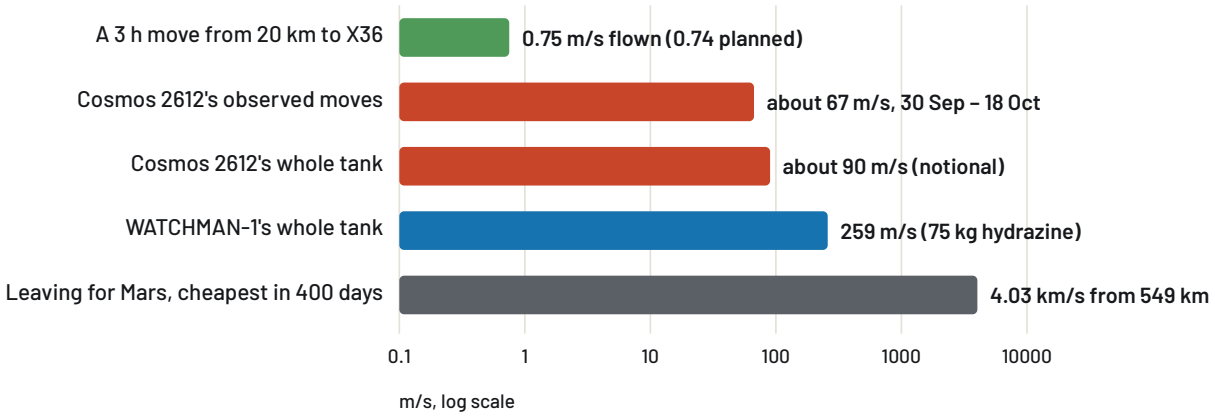
The planner shows a porkchop plot: departure wait against time of flight, each cell a refined transfer coloured by the Δv to go and the Δv to match on arrival. Its grid steps are design values. Close to a target, the

planner also offers station-keeping: holding a point a set distance ahead, behind, above or beside the target. A point 1 km out of ICEYE-X36's plane is held at 1.00 km.

3.5 Leaving Earth orbit

Where a scenario allows it, the planner plans between planets with patched conics (Curtis ch. 8; Vallado ch. 12). A heliocentric Lambert arc runs between the planets; the escape burn is fitted so its hyperbola leaves from the ship's actual position; and the plan is refined under full gravity. Spheres of influence follow Laplace, $r = a(m/M)^{2/5}$ (Curtis §8.4). From ICEYE-X36's orbit, the cheapest departure for Mars in the 400 days after the start costs 4.03 km/s. WATCHMAN-1 carries 259 m/s, Cosmos 2612 about 90. Interplanetary flight is switched off in CLOSE PASS, and these figures show it would be out of reach anyway.

FIG. 4 FUEL AGAINST WHAT THINGS COST



Δv from §3.3–3.5 and §9. Close-in moves are cheap; WATCHMAN-1 can make hundreds of them. Leaving Earth orbit is out of reach of every craft here, which is real physics, not a game limit.

4 RADIO AND RADAR: ANTENNAS, LINKS AND DETECTION

4.1 Antenna gain and patterns

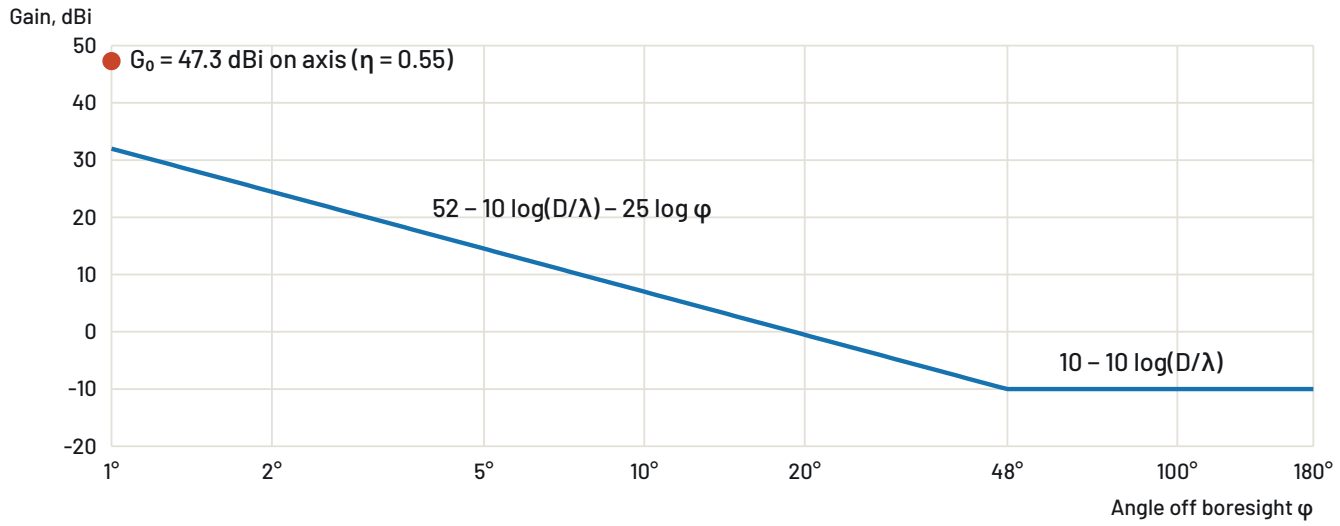
The peak gain of a dish of diameter D at wavelength λ is $G_0 = \eta(\pi D/\lambda)^2$, with $\eta = 0.55$ throughout. That is 1.24 dB conservative against the DSN 34 m antenna at X band (68.24 dBi at 8425 MHz, DSN 810-005 Module 104 Rev. O Table 8[39]). The beamwidth is about $70\lambda/D$ degrees (NAWCWD TP 8347 §3-1[54]; ITU-R SA.509-3 §1.3[33]) V-TEXT.

Off-axis gain follows the ITU reference envelopes: SA.509-3 for large dishes (D/λ ≥ 100), and S.465-5 Note 4 for small ones[34]:

$$G(\phi) = 52 - 10\log(D/\lambda) - 25\log\phi \quad \text{dBi}, \quad 100\lambda/D \le \phi < 48^\circ$$
$$G(\phi) = 10 - 10\log(D/\lambda) \quad \text{dBi}, \quad 48^\circ \le \phi \le 180^\circ$$

These are envelopes of sidelobe peaks, used in interference coordination, so they are conservative for a well-designed antenna. Note 2 of S.465 warns against using them below D/λ = 50, and every Red dish in the game is that small. They are still the most credible public pattern. Every emitter points. In CLOSE PASS, Cosmos 2612's command radio aims its 0.3 m dish at its handler near Moscow, and WATCHMAN-1 listens through a 0.5 m dish of its own. Warships and buoys in the engine's other world listen through four-face arrays that form beams in all directions (after the TDRS S-band multiple-access array, NASA TM-105422), with gain equal to the face's peak plus 10 log cos(scan).

FIG. 5 REFERENCE ANTENNA ENVELOPE



ITU-R SA.509-3 sidelobe envelope drawn for an illustrative dish 100 wavelengths across. It is an envelope of sidelobe peaks, so it is conservative for a well-designed antenna.

4.2 Link budget and noise

$$P_r = P_t\, G_t(\theta_e)\, G_r(\theta_r)\, (\lambda / 4\pi R)^2 \qquad \text{(Friis 1946)}$$
$$N = k\, T_s\, B$$

Free-space loss is the exact 20 log(4πR/λ) of ITU-R P.525-4 eq. (3). The practical eq. (4) rounds its constant to 32.4 dB (exactly 32.448 dB), so the game differs from eq. (4) by 0.048 dB.

4.3 Detection of unknown bursts

Listeners don't know a burst's waveform, so they detect it with an energy (radiometer) detector (Urkowitz 1967[51]). The integrated SNR grows as $\sqrt{(B\tau)}$. The chance of detection is:

$$P_d = Q\left(\left(Q^{-1}(P_{fa}) - d\right) / \sqrt{(1 + 2s)} \right), \quad P_{fa} = 10^{-6}$$

The false-alarm rate is a DESIGN VALUE. This expression matches the exact chi-square distributions to 0.01 for $n \geq 10^5$ samples, and even odds come at 6.8 dB. Spreading a signal over a wider band costs an eavesdropper's energy detector $\sqrt{(B_{old}/B_{new})}$ in SNR at fixed power (the radiometer equation; Urkowitz 1967[51]). ITU-R SM.1055[38] §1 notes that spread signals are noise-like and not readily detected.

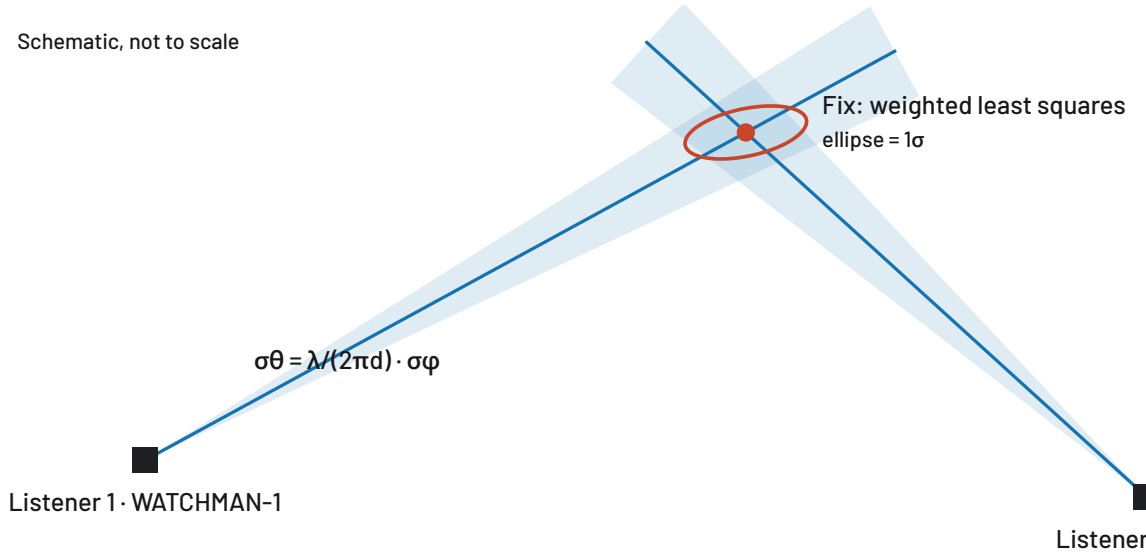
4.4 Direction finding and identification

A two-element interferometer measures a bearing with

$$\sigma_\theta = \lambda/(2\pi d) \cdot \sigma_\phi, \quad \sigma_\phi^2 = (1 + 2s) / (2B\tau\, s^2)$$

This is the cross-correlation phase error for a noise-like signal. It follows from the Knapp & Carter (1976) Cramér–Rao bound in terms of coherence, $\gamma^2 = s^2/(1 + s)^2$, in the narrowband limit; it is checked by a Monte Carlo of a two-element correlator to 3%. A 20 µrad floor is a DESIGN VALUE. Bearings from separated listeners are crossed by weighted least squares. Identification compares a recording's modulation and symbol rate against a threat library above a minimum SNR, a design-value gate. Specific-emitter identification is real: it recognises a device by imperfections in its analog circuitry (Danev et al.[78]).

FIG. 6 DIRECTION FINDING



Two listeners (in CLOSE PASS, WATCHMAN-1's two receivers 2 m apart, or two craft) each take a bearing with its own error wedge. The fix is their weighted least-squares crossing; the farther apart the listeners, the tighter the ellipse. Schematic.

4.5 Ground radar

Five real radar sites track the three craft. Each look returns range, azimuth and elevation, each with Gaussian noise, every 10 seconds while the craft is above the site's elevation mask and inside its reach. The site's position follows the turning Earth (§2.4): PFISR, at 65.1° N, moves 704 km in an hour, at 0.1962 km/s: the Earth's rate times its 2,690 km distance from the polar axis on WGS 84 gives 0.1961 km/s (DERIVED). Range noise is published for three of the sites; angle noise and masks are notional.

SITE	OPERATOR	RANGE Σ	ANGLE Σ	MASK	REACH	SOURCE
PFISR, Poker Flat, Alaska	SRI International / NSF; LeoLabs	14 m	0.05°	45°	3,000 km	Site and range noise: Nicolls et al. 2017[171], Table 1
Midland Space Radar, Texas	LeoLabs	14 m	0.1°	30°	3,000 km	Site and range noise: Nicolls et al. 2017[171]
Kiwi Space Radar, New Zealand	LeoLabs	17 m	0.02°	40°	1,500 km	Site, range noise and reach: Stevenson et al. 2020[172]
Arizona Space Radar	LeoLabs	15 m	0.05°	20°	3,000 km	Site: FCC experimental licence WO2XME[173]; noise notional
Space Fence, Kwajalein	U.S. Space Force, 20 SPCS	15 m	0.02°	20°	3,000 km	Notional: not published

Table 2. Ground radar in CLOSE PASS. Angle noise, masks and Space Fence's figures are DESIGN VALUES.

A radar at a site differs from one on a ship in two ways that matter here. It sees a craft only for a few minutes per pass, so custody between passes rests on the filter's prediction (§7). And its range noise, tens of metres, is far smaller than its angle noise at 1,000 km (0.05° is 870 m across the line of sight), so a single pass fixes how far away a craft is much better than where across the sky.

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5 ELECTROMAGNETIC ATTACK AND TENSION

The engine models jamming, spoofing, dazzling, high-power microwaves, cyber takeover and kinetic kills, and its other world uses them all. CLOSE PASS gives Blue none of them: WATCHMAN-1's only lever is its presence. The models are kept here because they are part of the engine, and because a scenario that adds a jammer would rely on them.

5.1 The jamming equation

Jamming a command uplink means being louder, at the victim's receiver, than its handler. Both paths are one-way, so the equation is Friis applied twice. This is the communications form of NAWCWD TP 8347 §4-9 V-TEXT, and Adamy's EW 101[75] gives it in dB form:

$$J = P_j + G_j(\text{aim error}) + G_v(\theta_j) - L_j$$

$$S = P_u + G_u + G_v(0) - L_s$$

$$J/S = J - S - 10\log(B_j/B_v) \quad \text{dB}$$

$G_v(\theta_j)$ is the victim's gain toward the jammer, at angle θ_j off its line to the handler, so geometry matters: a jammer near the handler's line faces the victim's main beam. The aim term means a loose track costs beam-pointing gain. Before the signal is identified, a jammer must spread its power across the whole allocation (barrage); once identified, it can put it all in the channel (spot). How much J/S breaks a link has no universal number. NAWCWD §4-7 (p. 4-7.3) says plainly that effectiveness "is not a direct mathematical function of J/S" V-TEXT. The engine requires +6 dB, a DESIGN VALUE inside the 0–10 dB band typical for FEC-coded BPSK.

5.2 Other effects

EFFECT	MODEL	SOURCE / STATUS
Spoof (FAKE)	False ranging tones capture the target's ranging loop at J/S ≥ 0 dB, inducing a drift	Loop capture at 0 dB and a drift of 900 km/h are design values
Dazzle	Spot radius $R\sqrt{((1.22\lambda/D)^2 + j^2)}$ at 1.06 µm, 2 m aperture, 0.2 µrad jitter; 10 W/m² at the aperture	Airy disc 1.22λ/D (Hecht); jitter added in quadrature (engineering approximation); 10 W/m² is an engineering estimate allowing for an OD-6 filter
HPM (PULSE)	Far field $E = \sqrt{(30PG)}/R$ from a 1 GW-peak X-band source; front-door damage at 100 V/m	$E = \sqrt{(30PG)}/R$ (Balanis; ITU-R P.525-4 eq. 1); 100 V/m is a DESIGN VALUE (engineering estimate); GW-class backward-wave oscillators (Benford et al. 2016)
Cyber (TAKE)	Takeover after recording enough command traffic	Abstracted. Real command links are encrypted, and this is stated as fiction
Kinetic kill	$P_k = \text{rel} \cdot (1 - e^{-r^2/2\sigma^2})$, from divert reach against track uncertainty	Rayleigh. Breakup by the NASA standard breakup model (below). The 1,200 catalogued pieces are a DESIGN VALUE set below FY-1C (2007; 3,442 catalogued[138]) and Cosmos 1408 (2021)

A kill breaks the wreck up by the NASA standard breakup model (Johnson et al. 2001[131], as given in Frey & Colombo 2021 App. B[135]). An impact is catastrophic above 40 J/g, and $N(\geq L_c) = 0.1\ M^{0.75}\ L_c^{-1.71}$ fragments are drawn. Each fragment's area-to-mass ratio $\chi = \log_{10}(A/M)$ follows App. B: the bimodal spacecraft law above 11 cm, the small-object law below 8 cm, and a linear blend in $\log L_c$ between 8 and 11 cm. Its speed follows $\log_{10} \Delta v \sim N(0.9\chi + 2.9, 0.4)$, Δv in m/s.

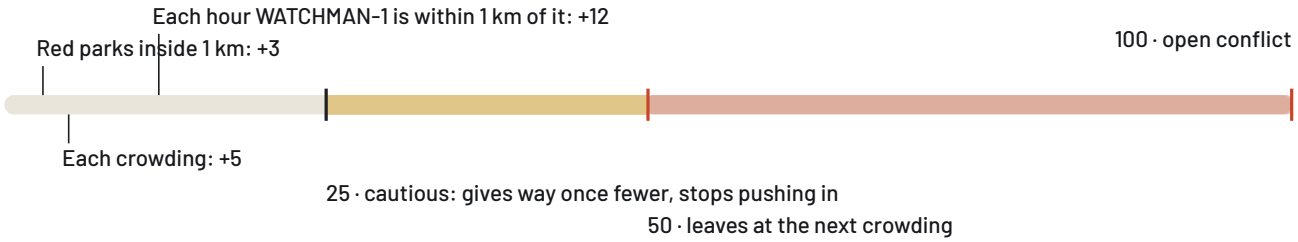
5.3 Escalation in the engine

Each effect carries an escalation cost, a design value: jamming 2 per hour, radar 1 per hour, spoofing 4, dazzling 8, cyber 10, high-power microwave 15, and a kinetic kill 45. Doctrine provides no escalation ladder (§8).

5.4 Tension in CLOSE PASS

CLOSE PASS has its own meter, because its only acts are approaches. Tension rises by 3 when Cosmos 2612 parks inside ICEYE-X36's 1 km keep-out, by 5 each time WATCHMAN-1 crowds it into giving way, and by 12 for every hour WATCHMAN-1 spends within 1 km of it. At 25, Cosmos 2612 turns cautious: it gives way once fewer before leaving, and it stops pushing in. At 50 it leaves at the next crowding. At 100 the confrontation has become open conflict and the scenario is lost. Every value is a DESIGN VALUE.

FIG. 7 TENSION IN CLOSE PASS



The tension meter of §5.4. Every value is a DESIGN VALUE. Doctrine gives no ladder (§8); this one rewards pressure that stops short of contact.

6 HEAT, INFRARED AND VISIBLE SIGNATURES

6.1 Energy balance

Each craft has a skin and, if it carries a reactor, radiators. Skin temperature follows a single-node balance with heat capacity (Gilmore[103]):

$$C\,\frac{dT}{dt} = P_{leak} + \alpha\, S\, f\, A/4 - \varepsilon\,\sigma\, T^4\, A, \qquad C = 897\ \mathrm{J\,kg^{-1}\,K^{-1}} \times 0.25 \times m_{dry}$$

$S = 1{,}361\ \mathrm{W\,m^{-2}}$ at 1 AU (Kopp & Lean 2011[109]), scaled by the inverse square of the Sun's distance, and f is the sunlit share (§6.4). The heat capacity is aluminium's at 298 K, 24.21 J $\mathrm{mol^{-1}\,K^{-1}}$ or 897 J $\mathrm{kg^{-1}\,K^{-1}}$ (NIST-JANAF, NIST Chemistry WebBook[105]); Incropera[104] Table A.1 gives 903 J $\mathrm{kg^{-1}\,K^{-1}}$ at 300 K, 0.7% more. The 25% skin fraction is a DESIGN VALUE. WATCHMAN-1's skin has a time constant of about 1.5 hours (DERIVED: $C = 897 \times 0.25 \times 600\ \mathrm{kg} = 134{,}550\ \mathrm{J/K}$ against $4\varepsilon\sigma T^3 A \approx 25\ \mathrm{W/K}$), so it cools noticeably during each pass through the Earth's shadow, which lasts up to about 36 minutes at 549 km. The Earth's own reflected sunlight (albedo

0.367) and its infrared glow (254 K as a black body) are not yet added to the heat balance (§11). Absorptance and emittance come from NASA RP-1121 (Henninger 1984[95]).

The engine also models reactor radiators, after the 100 kWe Brayton plant of NASA Glenn's JIMO study (Mason 2003[96]; Siamidis 2005[98]), as two-sided Lambertian plates, $I(\theta) = \varepsilon\sigma T^4 A \cos\theta/\pi$. No craft in CLOSE PASS carries a reactor.

6.2 Going dark

A reactor ship can stop radiating for a while by storing its heat in water it vents (2.5 MJ/kg) or ice it melts (333 kJ/kg, Leimkuehler 2011[102]). This is part of the engine's other world; CLOSE PASS has no use for it.

6.3 In-band infrared detection

Telescopes work in NEO Surveyor's NC2 band, 6–10 μm (Mainzer et al. 2023[115] V-TEXT). The in-band share of each body's emission comes from the Widger & Woodall series, which matches astropy to 6×10^{-11}. The 0.5 m LWIR's 1σ noise-equivalent irradiance is 7×10^{-18} W m^{-2}. That lies inside the 0.4–1.1×10^{-17} range derived from NEO Surveyor's S/N-5 sensitivity. A look detects with

```
S/N = I / R² / (NEI × background),     P_d = Φ(S/N − 5)
```

Against the planet's disc, the background raises the NEI ×50, a DESIGN VALUE inside the ×26–92 derived for a background-limited sensor. Just off the disc, stray light raises it ×5, also a design value, out to three times the planet's apparent radius but never more than 20° past its limb. From low orbit the Earth fills almost half the sky, and three times its apparent radius would leave no sky at all; 20° is about the Hubble Space Telescope's bright-Earth avoidance angle. False alarms are Poisson, with a mean of 0.1 per sensor per look (design value), and only lines from two places at once start a track.

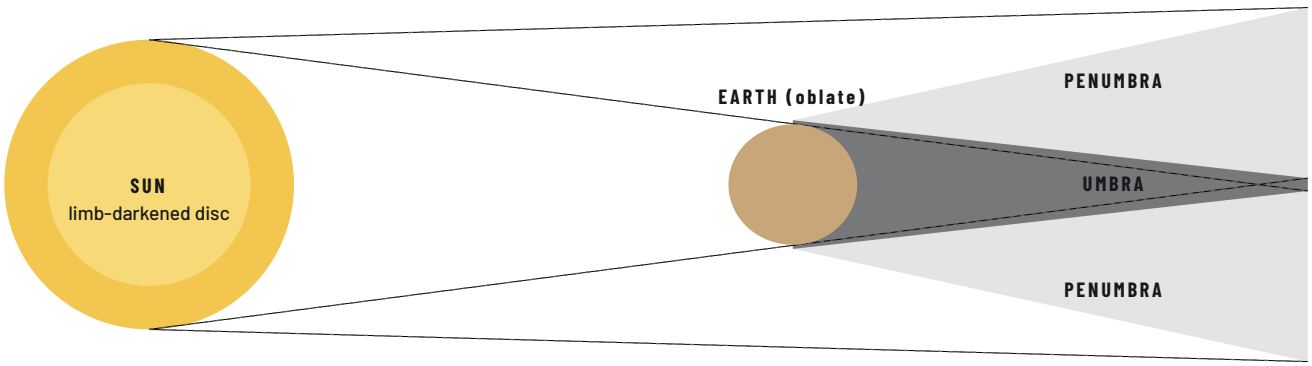
6.4 Shadows and reflected light

The Sun is a limb-darkened disc, $I(\mu) = 1 - u(1 - \mu)$ with $u = 0.6$ (Allen's[106] §14.7), sampled by 397 rays in 12 rings. The sunlit share is the weighted fraction not blocked by the oblate Earth (6,378 × 6,357 km), the Moon, or, far out, a planet. This gives both umbra and penumbra (Montenbruck & Gill §3.4.2), within 0.019 of a dense numerical integration.

Sunlit hulls are also seen in visible light. The Sun's $M_V = 4.81$ (Willmer 2018) is scaled to the irradiance at the craft, and each hull is a Lambert sphere with phase law $\Phi(\alpha) = (\sin\alpha + (\pi - \alpha)\cos\alpha)/\pi$. It returns a share $p\cdot\Phi(\alpha)\cdot r^2/R^2$ of the sunlight at its distance R, with p the geometric albedo and $r^2 = A/4\pi$ for a hull of surface area A (Madhusudhan & Burrows 2012 eq. 3[122]). An earlier build used the cross-section πr^2 in place of r^2, which made every hull π times (1.24 mag) too bright and every visible range $\sqrt{\pi} = 1.77$ times too long. A 1 m

GEODSS-class telescope reaches magnitude 22 (Stokes et al. 1996[119]; the free 1998 Lincoln Laboratory Journal account of the same tests gives magnitude 22 at S/N 4 in under 100 s). The game asks for S/N 5 at magnitude 22, a DESIGN VALUE. Visible telescopes look only where a track already exists, because reaching magnitude 22 takes about 100 s staring at one field.

FIG. 8 UMBRA AND PENUMBRA



Schematic, not to scale. 397 rays in 12 rings sample the disc; the share not blocked is the sunlit fraction f.

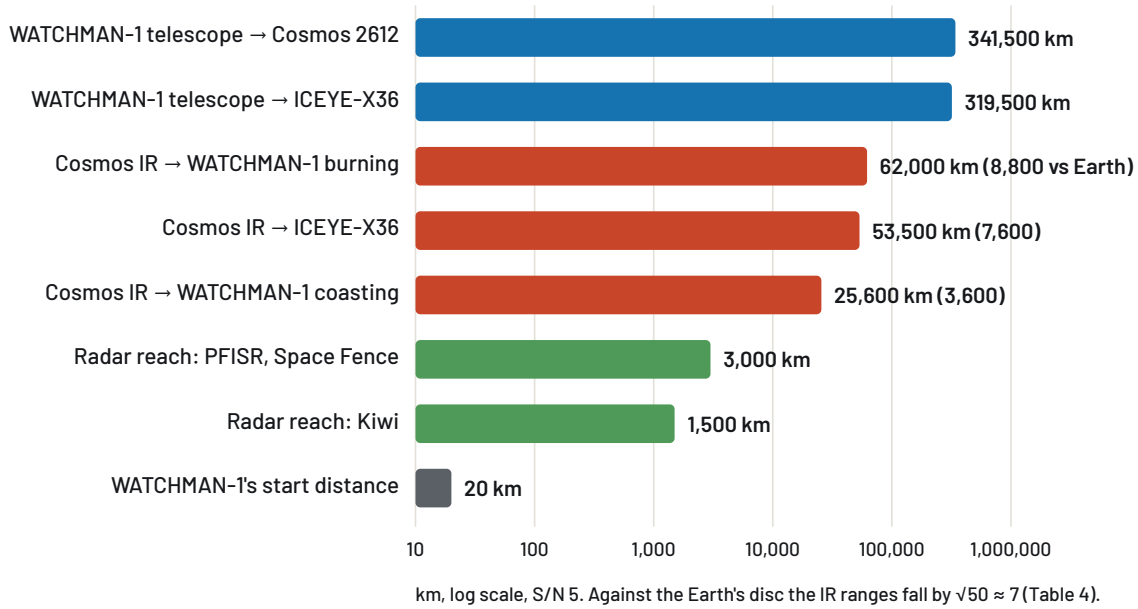
Because the Sun is a disc, the Earth casts a full shadow inside a soft-edged one (§6.4). At 549 km ICEYE-X36 spends up to about 36 minutes of each 95.6-minute orbit in it, depending on the season and its orbit's angle to the Sun.

6.5 What the scenario's sensors see

OBSERVER → TARGET	BAND	AGAINST DEEP SKY	AGAINST THE EARTH
WATCHMAN-1's 0.3 m telescope → Cosmos 2612	Visible, sunlit	341,528	–
WATCHMAN-1's telescope → ICEYE-X36	Visible, sunlit	319,470	–
Cosmos 2612's 0.1 m camera → WATCHMAN-1, coasting	6–10 µm	25,611	3,622
Cosmos 2612's camera → WATCHMAN-1, burning	6–10 µm	61,962	8,763
Cosmos 2612's camera → ICEYE-X36	6–10 µm	53,518	7,569

Table 4. Detection ranges at S/N 5, km, at the start (S = 1,361 W m^{-2}), measured from the engine's own functions. Against the Earth's disc the NEI is ×50, so infrared ranges fall by √50. The telescope looks only where a track already exists (cued), and only at sunlit targets.

FIG. 9 WHO CAN SEE WHOM, AND HOW FAR



From Table 4 and §4.5. Every sensor here outranges the scenario by a wide margin: what limits seeing in CLOSE PASS is geometry (sunlight, the Earth in the way, a radar site's pass), not range.

For the visible ranges the hull is a Lambert sphere (above). With a 1 m telescope the same hulls are seen at 583,270 km (ICEYE-X36), 623,542 km (Cosmos 2612) and 1,150,225 km (WATCHMAN-1), each $\sqrt{(1/0.3)} = 1.83$ times the 0.3 m range, because the limiting flux scales with aperture and flux falls as the square of range. These are long because the Sun at 1 AU is 27 times brighter than at Jupiter.

7 TRACKING

Every contact is a track: an estimate with a covariance, not ground truth. Bearings and ground radar looks (range, azimuth and elevation, §4.5) update it in three dimensions through an unscented Kalman filter (Julier sigma points, $\kappa = 0$, circular mean for angles), which matches filterpy to 3×10^{-12}. Prediction uses central-difference state transition matrices and white-noise acceleration (Bar-Shalom et al. 2001). Two cases needed special handling.

- **Close in.** With WATCHMAN-1 a few kilometres from Cosmos 2612, sigma points can straddle the sensor. Within a tenth of the range, a bearing therefore enters as a pseudo-linear measurement (Lindgren & Gong 1978; Aidala & Nardone 1982). Between craft in nearly the same orbit, range is unobservable from angles alone without a manoeuvre (Woffinden & Geller 2009). So without a recent radar range, the track keeps at least 0.5 × range of 1σ along the line of sight (design value).

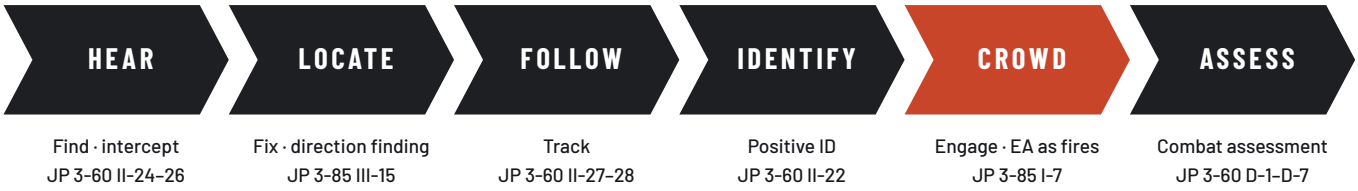
- **Unseen burns.** A bearing whose normalised innovation passes the $\chi^2(1)$ 99.9% point (10.83) is taken as a burn nobody saw, and widens the track (Bar-Shalom ch. 11). The 50 m/s and 10 min used are design values.

CLOSE PASS scores *custody*: Blue's track of Cosmos 2612 within 1 km of the truth for at least 80% of the time (design values). Between radar passes the track coasts on the force model of §3, so custody depends on how well the filter predicts. In the unattended 24-hour run of Fig. 12, with no orders and only the radar sites looking, custody held 94% of the time. Tracks also size the burns they see: a jump in a track's motion is reported as a burn of about so many m/s, give or take its uncertainty.

A track nobody has detected for 24 h, or whose 1σ exceeds 100,000 km, is dropped (after Blackman & Popoli 1999). Both thresholds are design values. Contacts are drawn after MIL-STD-2525D, with a dashed frame for a last known position.

8 DOCTRINE AND THE KILL CHAIN

FIG. 10 THE GAME'S VERBS AS A KILL CHAIN



Each step mapped to its doctrinal term and citation (table below).

The game's verbs follow US joint doctrine, read from the published texts: JP 3-85 *Joint Electromagnetic Spectrum Operations* (2020)[139], JP 3-60 *Joint Targeting* (2018)[141], AFDP 3-60 (2026)[142] and the USSF *Space Warfighting* framework (2025)[143].

GAME STEP	DOCTRINAL TERM	CITATION
Hear	F2T2EA *Find*; electromagnetic support search and intercept	JP 3-60 II-24–26; JP 3-85 I-6
Locate	*Fix*; direction finding, precision geolocation	JP 3-60 II-26–27; JP 3-85 III-15, B-4
Follow	*Track*	JP 3-60 II-27–28
Identify	Positive identification by EW support systems or visual recognition (doctrine acquires it during *Fix*)	JP 3-60 II-22
Crowd	No doctrinal term: a show of presence by an approach that stops short of contact. In the engine's other world this step is *Jam*, doctrine's *Engage* by electromagnetic attack (JP 3-85 I-7, I-13, GL-6, GL-7)	—
Assess	Combat assessment, with confidence graded confirmed, probable or possible	JP 3-60 D-1–D-7

Under JP 3-85, a stealth ship going dark is practising electromagnetic protection: "multispectral low-observability" is one of its listed examples. Doctrine says nothing about TDOA/FDOA, look-through, own-platform jammer isolation, or an escalation ladder. JP 3-85, JP 3-60 and AFDP 3-60 never use the words "spoofing" or "reversible": their terms are intrusion and deception, and temporary or nonlethal effects. (The USSF framework does speak of reversible and nonreversible actions, pp. 5 and 9.) Nor does it treat crowding another craft as an act at all. The tension meter of §5.4 is therefore entirely a design value. Its ordering (reversible and deniable acts cheap, debris-generating acts ruinous) follows doctrine's escalation-of-force guidance qualitatively.

9 THE SCENARIO AND ITS REAL-WORLD BASIS

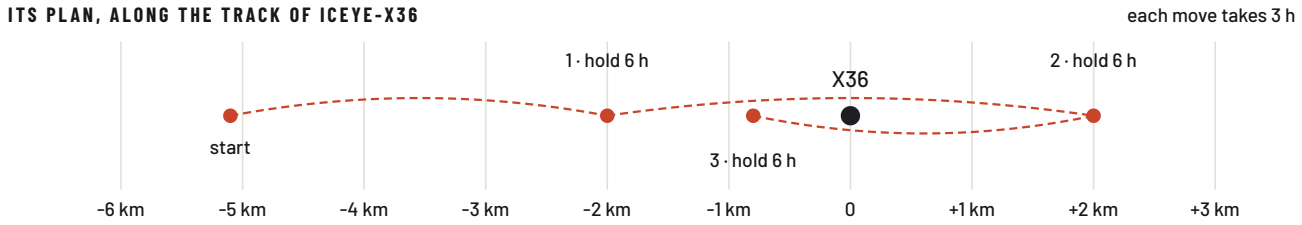
CLOSE PASS is built on a real uncooperative rendezvous and proximity operation. In October 2026 the Russian satellite Cosmos 2612 approached ICEYE-X36, a commercial X-band radar-imaging satellite, and shadowed it at a few kilometres. It was reported from LeoLabs tracking data by Breaking Defense and The War Zone on 19 October 2026[158,159], and reconstructed from public element sets by KeepTrack[168]. As reported, the sequence was: coplanar alignment; closing from long range; station-keeping within 5 km, then 1–2 km; a closest pass reported at 500 m (another report gives 26 m); and moves ahead of and behind the target. The scenario keeps the real craft and their real orbits. WATCHMAN-1, the US inspector the player commands, is notional, and so is everything Cosmos 2612 does after the start.

- **Orbit.** 549 km, a 95.57-minute period, 7.584 km/s (§2.2). Each orbit takes the craft through the Earth's shadow and over different radar sites.
- **Cosmos 2612's plan.** In the order LeoLabs reported the real one: close in from behind to 2 km, reposition 2 km ahead, then loiter 0.8 km behind. Each move takes 3 hours and each slot is held for 6. Distances and

times are DESIGN VALUES.

- **Coming closer.** When no Blue craft is within 50 km of it, it takes its next slot at half the distance, but not under 0.5 km.
- **Giving way.** With a Blue craft within 5 km of it, it gives way to a slot 8 km on the far side of ICEYE-X36. Crowded again after two give-ways (one in the tutorial), it leaves: 4 m/s prograde, and gone once it is beyond 200 km.
- **Its radio.** An S-band (2.25 GHz) BPSK burst at 100 kbaud, 5 W into a 0.3 m dish aimed at its handler near Moscow (the Titov control centre is at Krasnoznamensk), two minutes long, every 3 to 6 hours and after each move. These are design values within ordinary practice.
- **Losing its link.** Cut off from its handler for 3 hours within 200 km of its target, and with a Blue craft within 25 km of it, it leaves. Cut off but alone, it holds its slot and waits. This is grounded in practice, not in a published rule: a Progress approach to the ISS was aborted on loss of its telemetry link in 2010[162], and ESA's ATV carried RETREAT and ESCAPE modes for the same reason. The threshold and the burn are DESIGN VALUES.
- **Score.** Making it break off counts 40; keeping it outside 1 km of ICEYE-X36, 20; custody, 20; using no more than half of WATCHMAN-1's fuel, 10; staying below open conflict, 10. Design values.
- **Timescale.** The real event took weeks. The scenario runs two days from 21:00 UTC on 26 October 2026 DESIGN VALUE.

FIG. 11 HOW COSMOS 2612 BEHAVES



WHAT CHANGES ITS PLAN

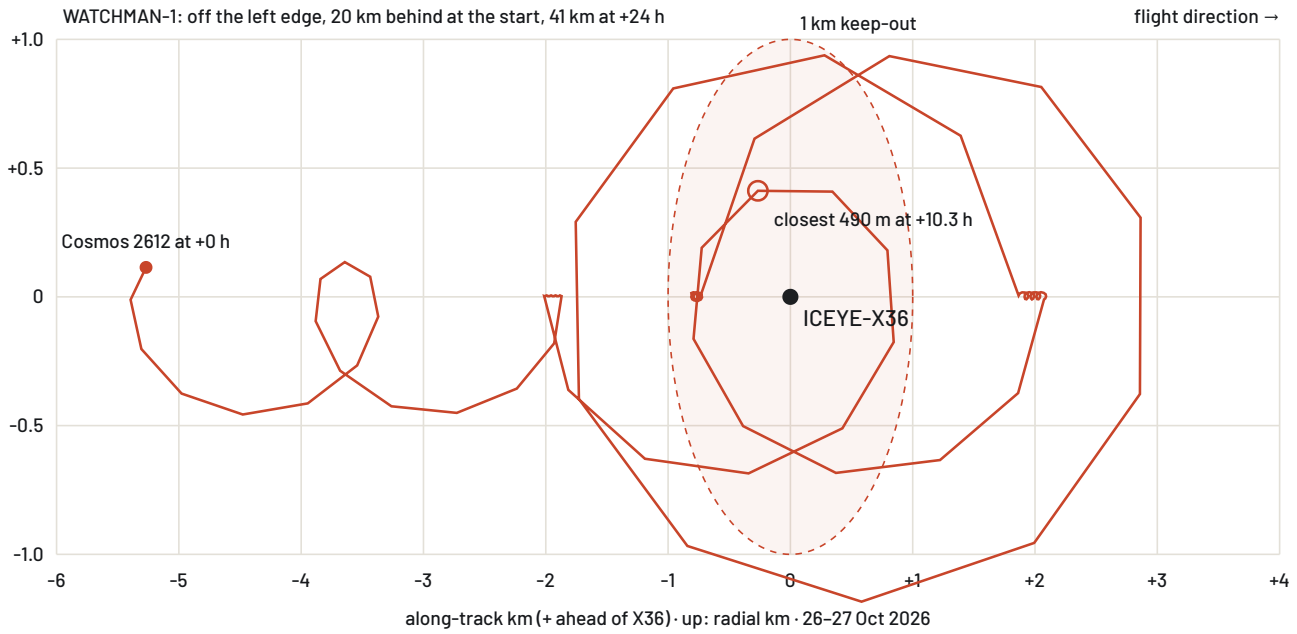
No Blue craft within 50 km	takes its next slot at half the distance (not under 0.5 km)
Blue within 5 km of it	gives way to a slot 8 km on the far side of X36
Crowded again after 2 give-ways	leaves: +4 m/s prograde, gone beyond 200 km
Cut off from its handler 3 h, Blue within 25 km	leaves; cut off but alone, it waits

Top: its plan along ICEYE-X36's track (§9). Bottom: what makes it come closer, give way or leave. Every threshold is a DESIGN VALUE.

Left alone, the plan runs to completion. Fig. 12 is a real 24-hour run of the engine with no orders given: Cosmos 2612 parks 2 km behind ICEYE-X36 at +3 h, passes within 490 m of it at +10.3 h on its way to its slot 2

km ahead (+11.5 h), and at +19.8 h parks 0.8 km behind, inside the keep-out. The keep-out objective fails, while WATCHMAN-1, given no orders, drifts from 20 km to 41 km behind.

FIG. 12 CLOSE PASS LEFT ALONE FOR 24 HOURS



A real Core run (`cli/run.gd --hours=24 --step=600`, seed 4417, no orders), plotted from its debrief: Cosmos 2612's position relative to ICEYE-X36 every 10 minutes. It works through its plan and comes within 490 m at +10.3 h, inside the 1 km keep-out. Blue's track held within 1 km 94% of the time.

10 VERIFICATION

A Godot script runs every test case through the engine's own functions and writes the results out. A Python suite (pytest) then compares them with independent references, each comparison carrying its own tolerance. JPL data is cached, so the suite runs offline. The suite exercises the engine on its own world, at Jupiter, where JPL Horizons gives an exact reference for every body; the models are the same ones CLOSE PASS runs at the Earth. On 1 November 2026 it made 681 comparisons across 64 tests; 680 fall within tolerance, and 4 further rows report known simplifications. The one that fails is the check of who sees whom: it still assumes the glare round the planet's limb reaches three times its apparent radius, and predates the engine's cap of 20° past the limb (§6.3). Recomputed with the cap, it agrees on every pair the planet does not hide. The check, not the engine, is out of date, and it will be corrected.

MODEL	REFERENCE	TOLERANCE	WORST RESULT
Kepler propagation (94 cases, e to 0.999 and hyperbolic)	hapsira farnocchia; scipy DOP853	1e-8	7.1e-10
Lambert (111 transfers)	hapsira izzo	1e-6	7.3e-15
Worked examples	Curtis Ex. 3.7, 5.2; Vallado Ex. 2-4	½ last digit	pass
Force model (J2, third bodies)	hapsira perturbations; gradient of the J2 potential	1e-9	8.5e-11
DOP853 integration, 1 day	scipy DOP853, rtol 1e-13	1 cm	2.8 mm
J2 secular rates	Vallado eqs. 9-41/42	1%	0.40%
Galilean moons, Sun and planets	JPL Horizons	per body	pass
Antenna patterns (468 cases)	ITU-R SA.509-3, S.465-5, coded from the text	1e-9 dB	1.4e-14
Free-space loss	ITU-R P.525-4 eq. (4), 32.4 dB constant	0.05 dB	0.048 dB
Radiometer gain $\sqrt{(B\tau)}$	Monte Carlo energy detector	6%	1.5%
Interferometer DF σ	Monte Carlo two-element correlator	8%	3%
Radar SNR and monopulse angle	Skolnik; Barton	1e-9; 0.2%	5.6e-15; 0.14%
Jammer J/S and reach	Adamy, independent dB form; brentq	1e-9 dB	5.7e-14
Radiators against JIMO	Siamidis 2005 design point	2%	1.1%
Shadow fraction	Dense numpy disc integration	0.02	0.019
Skin heat capacity	scipy solve_ivp	0.05 K	0.0001 K
Unscented update	filterpy UKF	1e-6	3.3e-12
Rocket equation	Tsiolkovsky, every ship	0.006 km/s	0.0032 km/s

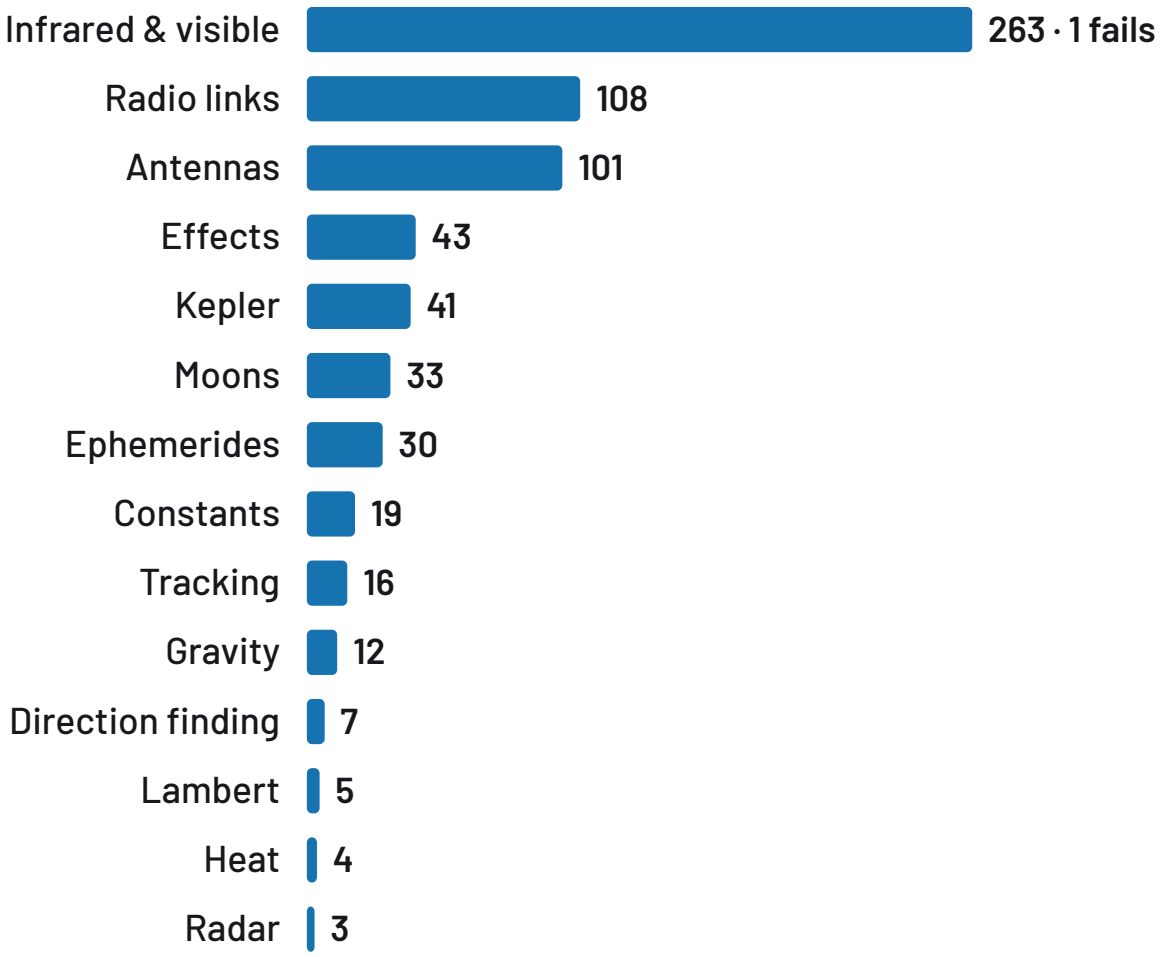
Table 5. Selected comparisons. The full table has one row per comparison.

The Earth's own numbers are checked by two Godot tests, `earth_test` and `close_pass_earth_test`, against textbook values. Both pass:

CHECK	ENGINE	REFERENCE
ICEYE-X36's node turns per day	1.0395°	1.0400°, J2's first-order secular rate (Vallado eqs. 9-41/42)
A craft on the no-drift path beside it, over one orbit	within 779 m	under 800 m: starting states and forces agree
The Sun's distance at the start	0.9938 AU	0.98–1.00 AU in late October
PFISR's speed on the turning Earth	0.1962 km/s	0.1961 km/s: ω × distance from the axis at 65.1° N on WGS 84
A station 1 km out of ICEYE-X36's plane	held at 1.00 km	1 km
A 3 h move from 20 km to ICEYE-X36	0.75 m/s flown	0.74 m/s planned
WATCHMAN-1's Δv	259 m/s	Tsiolkovsky: 2.2 km/s × ln(675/600)

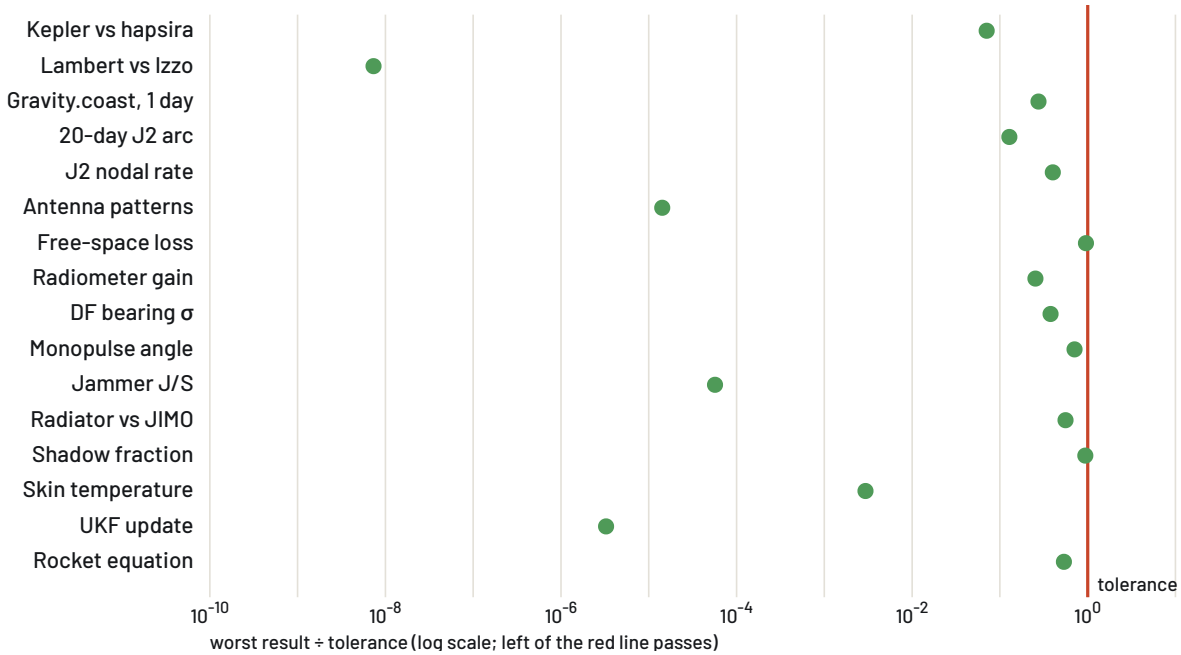
Table 5b. Checks of CLOSE PASS at the Earth.

FIG. 13 **WHERE THE 685 ROWS GO**



Rows per domain on 1 November 2026: 680 within tolerance, 1 out of date (§10), and 4 that report known simplifications.

FIG. 14 **HOW MUCH MARGIN EACH CHECK HAS**



Worst result divided by its tolerance, for the numeric rows of Table 5. Every dot sits left of the red line; most sit orders of magnitude inside it.

The suite has also found and fixed real errors. Moon longitudes were up to 114° off. DF σ was three times too large, because it was fed the integrated SNR. The barrage band was twice as wide as the real allocation. Long-way Lambert transfers failed. Two hulls were π times too bright in visible light. Kepler's equation lost accuracy near e = 1. Each fix and its effect on play is recorded in the project's verification log.

11 LIMITS, SIMPLIFICATIONS AND OPEN QUESTIONS

Not public, so notional

- Cosmos 2612's mass, engine, propellant, camera and radio. Only its orbit and its observed moves are public.
- WATCHMAN-1 as a whole. GSSAP's mass is reported; its sizes, thrusters and sensors were never published.
- Angle noise and elevation masks of the radar sites, and all of Space Fence's figures.
- The site and mask of Cosmos 2612's handler.

Not verified

- The closest approach of the real event: one report gives 500 m, another 26 m.
- HawkEye 360 geolocation accuracy; the Albersheim validity limits; specific-emitter identification accuracy; look-through duty cycles.
- The 2024 edition of JP 3-60 (the 2018 edition was read), and JP 5-0's definitions of MOE and MOP.
- MIL-STD-2525D was read through a reproduction of its tables, not the original PDF.

Simplifications

- No atmospheric drag. At 549 km it slows a small satellite by metres per second per year, depending on the Sun's activity; over two days it moves the three craft relative to each other by an amount that depends on their area-to-mass ratios, which are not public.
- No reflected sunlight or infrared from the Earth in the heat balance, though the Earth's disc does raise the infrared background (§6.3).
- A low-precision Moon (a few arcminutes) and an approximate Sun; no Earth gravity terms beyond J2.
- No nutation, polar motion or UT1 − UTC: ground sites sit within about 1 km of their true places.
- SGP4 is used only for the starting state; nothing is fitted back to element sets during play.
- The umpire's view: some kill-chain indicators still read ground truth rather than Blue's assessed belief.

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12 xGEO CORE AS AN OPEN-SOURCE TOOL

The models in this edition make up **xGEO Core**, the engine under the game. **xGEO Live**, the game, draws what Core computes and sends orders into it, and nothing more. Core runs with no window, and it is being prepared for release as an open-source simulation tool for teaching, concept studies and research. It is not a validated operational tool.

12.1 What is Core and what is Live

PART	WHAT IT HOLDS	WHERE
Core: physical models	Two-body motion, Lambert, gravity with J2 and third bodies, ephemerides, SGP4 starting states, the turning Earth and ground sites, link budgets, antenna patterns, radar, heat and infrared, the heat forecast, sensing, the tracking filter, breakup and debris, civil traffic, and the world that ties them together	`godot/core/`
Core: scenario rules	Blue's effects and their costs, Red's behaviour, objectives and the debrief, the log's wording, save and continue. They run headless, but they are rules of play, and most of their values are Notional	`godot/core/`
Scenarios	Classes, equipment and constants; CLOSE PASS's craft, radar sites, Red's plan and objectives	`godot/data/*.json`
Live	The console, the map, the windows, art, sound, the tutorial	`godot/ui/`, `godot/scenes/`, `godot/assets/`

Table 6. The split. A check in the test suite fails if anything in Core names one of Live's classes.

12.2 Running Core without the game

A command-line runner loads a scenario, steps time and writes what happens as JSON lines, one object per line:

```
godot --headless --path godot --script res://cli/run.gd -- \
      --hours=24 --step=600 --seed=4417 --out=run.jsonl
```

The first line describes the run (scenario, seed, start time, units). Each step line carries the time, escalation, Blue's ships (position, velocity and Δv left), Blue's tracks with their 1σ ellipses, and every log entry since the last step. The last line is the debrief. Tracks report Blue's belief, never the truth; `--truth=true` adds the true states for analysis. Units are kilometres, km/s and seconds, centred on the scenario's central body: the Earth in CLOSE PASS. Run with no scenario, Core loads its own sample world at Jupiter. The test suite runs it on every change.

12.3 Real, Projected and Notional

Every model and value carries one of three labels, which extend the source labels of §1:

- **Real:** sourced and checked against an independent reference (§10). The orbits, gravity, the radio and radar equations, Planck's law, the filter.

- **Projected:** near-future and grounded in programmes that exist today, such as nuclear-thermal drives at NASA's tested 900 s specific impulse.
- **Notional:** invented for a scenario, the way defence and industry studies fly notional systems: Red's classes and behaviour, equipment sizes, escalation scores, the compressed timescale.

A Real model fed Notional inputs gives a Notional answer: the jamming equation is exact, but a Red jammer's power is a design value, and so is the range it yields. The project's fidelity table (`docs/FIDELITY.md`) gives each model's label, what it was checked against and what it leaves out.

12.4 Status of the release

Licence (26 October 2026): xGEO Core, the code of the models, the runner, the tests and the verification, is released under the Apache License 2.0. Anyone may use it and change their own copy for any purpose, keeping the notices and not calling their version xGEO. The game, its art and story, and this document stay all rights reserved. Core is open source but not open to contributions: bug reports, physics corrections and feature requests are welcome as issues; pull requests are not accepted. Done: the asset licence audit, the Core and Live split with its check, the command-line runner, the fidelity table, and contribution, conduct and security guides. The jamming and counterspace models are built only from open, published sources, every one cited in this edition, and the owner has cleared them and the name for release. CLOSE PASS names real satellites, operators and radar sites from public records; WATCHMAN-1 and everything the scenario has them do after its start are notional. Still to do before release: a review of the repository's history, then making it public. Issues must cite public sources only.

REFERENCES

Every source the game cites, in its code, data and tests, grouped by the model that uses it. A link opens the document itself, or a free copy where the publisher's is behind a paywall. Works with no readable copy online are listed without one.

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